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"Vegetation and Energy"
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VALIDATION REPORT - UPDATE 2025

LAND SURFACE TEMPERATURE

VERSION 3.0

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List of Acronyms

AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
CAL/VAL	Calibration / Validation
CCI	(ESA) Climate Change Initiative
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
CSV	Comma-Separated Values
DC	Diurnal Cycle
ECV	Essential Climate Variable
ESA	European Space Agency
GBOV	Copernicus Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary
GOES	Geostationary Operational Environmental Satellite
HMWR	Himawari
IODC	Indian Ocean Data Coverage mission
INSPIRE	Infrastructure for Spatial Information in the European Community
IPMA	Instituto Português do Mar e da Atmosfera
JRC	Joint Research Council
LSA-SAF	Satellite Application Facility on Land Surface Analysis
LST	Land Surface Temperature
LST10	Land Surface Temperature-10 daily composite
LPV	Land Product Validation
MSG	Meteosat Second Generation
NetCDF	Network Common Data Format
NRT	Near-Real Time
PQAD	Product Quality Assurance Document
PROBA-V	Project for On-Board Autonomy - Vegetation
PUM	Product User Manual
RM	Reference Measurements
RMSD	Root Mean-Squared Difference
S2GM	Sentinel-2 Global Mosaic
SKT	Skin Temperature
SYN	Synthesis
TCI	Thermal Condition Index
VR	Validation Report
WGS	World Geodetic System

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time-series.

Quality Assessment, and continuous Quality Monitoring constitute the only means of guaranteeing the compliance of generated products with requirements:

- The former concerns the new products/version which must pass an exhaustive scientific evaluation before to be implemented operationally.
- The latter concerns the operational products to check their quality keeps the same level along the time.

Both follow, as much as possible, the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products.

This document reports on the continuous quality assessment and validation of the CLMS global Land Surface Temperature (LST) version 3.0 for the operationally distributed family of products for the year 2025, including: LST hourly, 10-daily LST daily cycle (LST10-DC), and 10-daily LST Synthesis (LST10-Syn). The validation is based in comparisons of the CLMS LST estimates with in-situ observations. This method allows to verify the product's stability, as well as to perform a qualitative comparison of obtained statistics for the 2025 data with those found in previous years, including those found in the respective main validation report [CGLOPS1_VR_LST-LST10-V3.0].

Time-series of the various CLMS LST products are compared to the corresponding estimates from 20 in-situ stations, covering a range of land covers and climate zones. The source of in-situ data used in this report is the Copernicus Ground-Based Observations for Validation (GBOV) portal, which provides a set of reference measurements for several land surface variables. It is worth noting that up to the production of this report, a reasonable collection of 2025 in-situ datasets were either not available through the GBOV portal or had been reported to contain processing inconsistencies, while other sites were found not appropriated for remote-sensed LST validation. The current selection relies exclusively on high-quality datasets from GBOV version 2.2 (September 2023 release), which included significant corrections to previously detected anomalies, reflecting continuous improvements in GBOV quality control procedures.

Results in this report confirm that version 3.0 continues to demonstrate the overall consistency and stability of the CLMS LST family of products. The LST hourly product shows an overall bias of about -0.72 K and root-mean-square-difference of 3.36 K, which are comparable to those found in

the main validation report for the full-period 2018–2024 (about -0.50 K and 3.40 K, respectively), despite the different number of matchup samples used. Conformity tests show that 50 % of satellite/observation LST matchups meet the threshold requirement (i.e., below or equal to 2 K), while 27 % meet the goal requirement (i.e., below or equal to 1 K). The same conclusions apply to the LST10 family of products, which were evaluated against the corresponding statistics derived from clear-sky only in-situ data.

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1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document provides a yearly update to the quality assessment and validation for both CLMS LST and LST10 products generated for the year 2025. Product quality is assessed by comparisons between time-series of the operational products and a set of available in-situ stations that follow the procedures described in the VR.

The main goal is to verify that the validation statistics remain consistent with those reported in the VR and to confirm that the CLMS LST products still meet the required specifications. It should be noted that, within the framework of CGLOPS-1, only GBOV in-situ processed LSTs are considered for the 2025 evaluation.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements, and the expected performance.
- Chapter 3 provides a quick description of the validation methodology and lists the in-situ stations used.
- Chapter 4 presents the validation results for LST-hourly, LST10-DC, and LST10-Synthesis, as well as a brief summary of the product's temporal stability.
- Chapter 5 summarizes the main conclusions.
- Chapter 6 lists references cited in the report.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Copernicus Global Land Service
CGLOPS1_ATBD_LST-V3.0	Algorithm Theoretical Basis Document for Land Surface Temperature Version 3.0.
CGLOPS1_ATBD_LST10-V3.0	Algorithm Theoretical Basis Document of the 10-daily Land Surface Temperature Version 3.0.
CGLOPS1_VR_LST-LST10-V3.0	Validation report of the LST Version 3.0 product.
CGLOPS1_PUM_LST-V3.0	Product User Manual summarizing all information about the hourly Land Surface Temperature Version 3.0 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

Available at: https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

GBOV-ATBD-RM2-RM8-RM9	Ground-Based Observations for Validation – Algorithm Theoretical Basis Document – Land Surface Temperature products

Available at: https://gbov.acri.fr/products

2 REQUIREMENTS

According to the applicable document [AD1], the LST product specific technical details and requirements are summarized below:

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude–longitude grid
Geographic coverage	Global

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Spatial resolution	5 km
Time span:	Hourly and 10-day period
Timeliness:	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period.
Uncertainty (2 sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data structure

Data coding[1] shall be compatible with the CLMS products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product.
- The per-pixel date of the individual measurement or the start-end dates of the period covered.
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the CLMS production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current CLMS component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Levels of uncertainty: the error estimates will account for the uncertainty of the algorithms themselves. Stability may relate to the existence of a trend between the product and reference ground measured or modelled data.

Validation of the products shall conform to at least the Committee on Earth Observation Satellite (CEOS) Land Product Validation (LPV) standards (https://lpvs.gsfc.nasa.gov/). Wherever appropriate, the bio-geophysical variables shall be validated and compared to CEOS Calibration/Validation (CAL/VAL) datasets and/or Ground-Based Observations for Validation (GBOV, https://land.copernicus.eu/en/products/GBOV) of CLMS when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

To ensure continuity with the current operation, the same biomes as already used in the CLMS component shall be used.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu.

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts.
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the CLMS component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary (GEO) and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary datasets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the CLMS component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 VALIDATION METHODOLOGY

The validation methodology is based on a direct comparison between hourly and 10-daily satellite-derived LST estimates and in-situ observations from the GBOV service. Comparisons between in-situ observations and CLMS LST estimates are performed at the satellite pixel closest to each station location, and only matchups with an absolute difference in acquisition time less than 3 minutes (Trigo et al., 2008) are considered valid, particularly for stations measuring at 1 or 5-minute frequency. For stations reporting at 30-minute intervals (e.g., in Australia), the nearest satellite timeslot is used. According to the product requirements (chapter 2), the resulting satellite/in-situ LST matchup time-series are evaluated at each station using a set of statistical metrics designed to quantify accuracy, precision, and associated error of the CLMS LST estimates towards in-situ, as well as conformity tests performed at the satellite's pixel level, as described in the following sections 3.2 and 3.3, respectively.

In addition to the direct validation against in-situ observations, the temporal stability of the CLMS LST version 3.0 product is also considered in the present report to ensure continuity and consistency of the dataset from 2018 to NRT. For the 2025 exercise, this assessment had already been addressed in the main validation report [CGLOPS1_VR_LST-LST10-V3.0] using the ERA5 skin temperature product. A concise summary of the main findings is provided in the results section, while the complete methodology and detailed analyses are available in the main validation report.

3.1 IN-SITU STATIONS

The validation of CLMS LST V3.0 is based on the most recent GBOV (Bai et al., 2019) reference measurements for LST datasets (version 2.2, September 2023 release). The GBOV service collects raw measurements from each ground station in whatever format the source data presents itself, and provides harmonized LST reference values derived according to the instrumentation available at each site. GBOV output files are distributed in a standardized comma-separated values (CSV) format, facilitating their integration and direct comparison with satellite-derived estimates. The methodologies used to derive in-situ LSTs for each station are described in detail in the GBOV product ATBD [GBOV-ATBD-RM2-RM8-RM9].

Given the high spatial and temporal variability of LST, direct validation against in-situ measurements is challenging. Therefore, an initial data curation is performed to select only robust GBOV sites suitable for satellite LST validation. This pre-selection consists in comprehensive quality-control and data availability checks, including four main criteria: (i) at least 6 months of valid observations for the reporting year of 2025; (ii) location within the satellite field of view; (iii) no instrumentation issues affecting local records; (iv) consistency with surface representativeness at the satellite pixel scale. The latter two criteria are assessed through a detailed inspection of raw in-situ time-series and examination of the station surrounding environment using high-resolution orography, land cover

maps, and Landsat-based LST retrievals, implemented via the Google Earth Engine algorithm developed by Ermida et al. (2020).

The LST in-situ 2025 datasets available for the analysis (Table 1) are composed by a group of 20 stations, and were directly extracted from the GBOV portal (https://gbov.land.copernicus.eu/data-access). It is worth noting that, for the present report, several in-situ consistent datasets that had been used in past assessments are not considered for the analysis. These either result from data unavailability or to datasets that were found to be affected by LST processing inconsistencies when performing the corresponding emissivity correction. In particular, for the latter, this affected stations *Manhattan*, *Des Moines*, *Williams*, *DukeForest*, and *Southern Great Plains*. Additionally, station *Table Mountain* is also not considered due to representativeness issues, since this station is located close to a mountain range.

Table 1 - List of GBOV stations used in this report. Land cover information was retrieved from http://maps.elie.ucl.ac.be/CCI/viewer/index.php, and Köppen-Geiger climate classification data was obtained from http://koeppen-geiger.vu-wien.ac.at/.

Name	Code	Latitude	Longitude	Elev. (m)	CCI Land Cover	Köppen climate classification
Bondville (USA)	BON	40.052° N	88.3731° W	229	11 (Cropland)	Temperate, humid, hot summer (Cfa)
Fort Peck (USA)	FPK	48.31° N	105.10° W	634	130 (Grassland)	Arid, steppe, cold (BSk)
Desert Rock (USA)	DRA	36.62° N	116.02° W	1006	120 (Shrubland)	Cold arid (BWk)
Rock Springs (USA)	PSU	40.72º N	77.93° W	376	11 (Cropland)	Temperate, humid, hot summer (Cfa)
Sioux Falls (USA)	SXF	43.73° N	96.62° W	473	11 (Cropland)	Cold, humid (Dfa), hot summer (Cfa)
Goodwin Creek (USA)	GWN	34.25° N	89.87° W	98	40 (Mixed Vegetation)	Temperate, humid, hot summer (Cfa)
Bilos (France)	BIL	44.49º N	0.96 º W	38	70 (Tree cover)	Marine west coast (cfb)
Grignon (France)	GRIG	48.84º N	1.95º E	137	11 (Herbaceous cover)	Temperate, warm summer (Cfb)
Payerne (Switzerland)	PAY	46.81° N	6.94° E	637	10 (Cropland)	Temperate, humid (Cfb)
Cabauw (The Netherlands)	CAB	51.97º N	4.93º E	0	130 (Grassland)	Temperate, humid (Cfb)
SanRossore2 (Italy)	SRO	43.73º N	10.29º E	12	70 (Tree cover)	Meditteranean, hot summer (Csa)
Renon (Italy)	REN	46.59º N	11.43º E	1740	70 (Tree cover)	Cold, cold summer (Dfc)
Maasmechelen (Belgium)	MAAS	50.98º N	5.63º E	85	120 (Closed Shrublands)	Temperate, warm summer (Cfb)
Brasschaat (Belgium)	BRAS	51.31º N	4.52º E	16	110 (Herbacious cover)	Temperate, warm summer (Cfb)

Name	Code	Latitude	Longitude	Elev. (m)	CCI Land Cover	Köppen climate classification
Gingin (Australia)	GIN	31.38° S	115.71° E	50	62 (Tree cover)	Warm temperate, hot dry summer (Csa)
Robson Creek (Australia)	ROB	17.12° S	145.63° E	680	50 (Tree cover)	Temperate, humid, hot summer (Cfa)
Calperum (Australia)	CAL	34.00° S	140.59° E	64	120 (Shrubland)	Arid, steppe, cold (BSk)
Yanco (Australia)	YAN	34.99° S	146.29° E	119	10 (Cropland)	Arid, steppe, cold (BSk)
Cumberland Plain (Australia)	CBL	33.62° S	150.72° E	24	120 (Shrubland)	Temperate, humid, hot summer (Cfa)
Tateno (Japan)	TAT	36.06° N	140.13° E	26	10 (Cropland)	Warm, humid, warm summer (Cfa)

3.2 STATISTICAL METRICS

The evaluation of satellite-derived LST estimates is carried out using a set of robust statistical metrics computed from the obtained satellite/in-situ matchup pairs, which are spatially and temporally synchronized. This includes the median error, i.e., the bias (μ), as means to quantify the accuracy:

$$\mu = median(LST_{sat} - LST_{obs}), \quad (1)$$

where LST_{sat} denotes the satellite-derived LST estimate, and LST_{obs} the corresponding in-situ observation. For the precision, the median of the absolute residuals (σ) is used:

$$\sigma = median(|(LST_{sat} - LST_{obs}) - \mu|). \quad (2)$$

To account for both magnitude and distribution of the errors, the root-mean-squared-difference (RMSD) is also applied:

$$RMSD = \sqrt{\frac{\sum(LST_{sat} - LST_{obs})^2}{N}}, \quad (3)$$

where N is the number of valid matchup samples. To quantify the uncertainty associated with sampling in the metrics calculations, 95 % confidence intervals are computed for each statistical metric. These intervals are obtained through a bootstrap resampling approach (with 1000 iterations) applied to the series of matchup data. This method provides a robust estimation of uncertainty while accounting for the inherent variability in the observations, having been used successfully in previous studies (e.g., Dutra et al., 2014).

3.3 CONFORMITY TESTS

The assessment of CLMS LST estimates is further extended by performing conformity tests through the use of the apparent error at each pixel *i* (ε_i), following the product specifications (section 2). This error is defined as:

$$\varepsilon_i = LST_{sat} - LST_{insitu}. \quad (4)$$

With Δ representing the maximum permissible error, the conformity status is determined according to the following criteria:

- Conform: when the absolute error is less than or equal to the maximum permissible error:

$$|\varepsilon_i| \leq \Delta\ ; \quad (5)$$

- Non-Conform: when the absolute error exceeds the maximum permissible error:

$$|\varepsilon_i| > \Delta\ ; \quad (6)$$

The conformity results are expressed as the percentage of conforming and non-conforming cases for each station, providing a complementary measure of compliance with the product accuracy requirements.

4 RESULTS

4.1 IN-SITU VALIDATION

4.1.1 LST hourly product

A summary of the statistical results obtained for each GBOV station is presented in Table 2, following the evaluation metrics described in section 3.2. Conformity tests indicate the percentage of matchups that conform to the threshold specifications (i.e., when $|\varepsilon_i| \leq 2$ K) and with the goal specifications (i.e., when $|\varepsilon_i| \leq 1$ K), as described in section 3.3. Among all stations, PSU and GWN exhibit the lowest biases (0.02 K and 0.11 K, respectively), both associated with a comparable low RMSD (3.39 K). In contrast, larger errors are observed at REN, CAL, ROB, and TAT with RMSD values of 4.63 K, 4.27 K, 4.18 K, and 4.07K, respectively. These discrepancies are likely associated with surface heterogeneity (e.g., vegetation cover variability) and representativeness issues at the satellite pixel scale. For instance, REN and ROB are surrounded by trees, CAL is located within shrubland, and TAT is characterized by mixed urban vegetation and buildings. Overall, considering all hourly matchups, the CLMS LST version 3.0 for 2025 shows an accuracy of approximately -0.72 K, precision of 1.78 K, and RMSD of 3.36 K. Compliance with the threshold and goal specifications reaches about 50 % and 27 % of the matchups, respectively.

It is worth noting that, although the present analysis covers only the year 2025 and a reduced number of stations, the obtained results remain consistent with those reported in the main validation report for the 2018–2024 period [CGLOPS1_VR_LST-LST10-V3.0]. From the latter, considering the same analysis methodology but using only version 3.0 matchups against in-situ observations, a precision of about 1.80 K and RMSD of 3.40 K were obtained, while a lower bias (about -0.50 K) was obtained relative to the present analysis. These differences should take into account the substantially larger number of matchup samples used in the previous validation exercise (835893 vs. 66748).

Table 2 – Statistical summary for the 20 GBOV stations included in the annual validation (2025) of the CLMS LST V3.0. For each station, the statistical metrics (bias: μ, absolute residuals: σ, root-mean-squared difference: RMSD) and corresponding number of matchups are presented. Conformity with the LST product specifications is evaluated according to the defined threshold and goal criteria (see section 3). Global summary statistics computed using all hourly LST matchups from all stations are provided in the last row, including the 95% confidence interval between brackets. LST values are expressed in Kelvin, while percentages are used for threshold and goal requirements.

Station code	Land Cover	μ [K]	σ [K]	RMSD [K]	Threshold [%]	Goal [%]	# points
BON	Cropland	-1.03	2.34	3.15	60	33	3480
FPK	Grassland	-1.00	1.23	2.89	67	39	3460
DRA	Shrubland	-2.10	0.80	2.56	47	19	5717
PSU	Cropland	0.02	1.37	2.39	68	37	2715
SXF	Cropland	-1.01	1.30	2.98	60	33	3686
GWN	Mixed Vegetation	0.11	1.44	2.39	64	36	4031
BIL	Tree cover	-2.03	1.68	3.47	31	12	2709
GRIG	Herbaceous cover	-0.22	2.00	3.19	50	30	2751
PAY	Cropland	-0.95	1.73	3.23	57	28	3194
CAB	Grassland	-0.61	1.20	2.61	67	41	3262
SRO	Tree cover	-1.39	2.04	3.25	34	14	1900
REN	Tree cover	1.16	1.80	4.63	51	31	1483
MAAS	Closed Shrublands	1.01	2.71	3.74	38	19	2498
BRAS	Herbacious cover	-1.03	1.87	3.43	41	19	2634
GIN	Tree cover	-0.70	1.33	3.32	57	29	3536
ROB	Tree cover	-0.99	1.62	4.18	55	32	2110
CAL	Shrubland	-0.52	2.17	4.27	43	23	5661
YAN	Cropland	1.29	2.20	3.59	41	24	3990
CBL	Shrubland	-1.44	1.63	3.63	41	17	4275
TAT	Cropland	1.90	3.05	4.07	34	16	3656
Total	-	**-0.719** (±0.023)	**1.779** (±0.018)	**3.359** (±0.025)	**50.331** (±0.386)	**26.509** (±0.337)	**66748**

The yearly evolution of each performance metric is presented in Figure 1. In addition to the 2025 statistics (last row), yearly results for 2018–2024 and the full 7-year period are included using the validation framework from the main validation report [CGLOPS1_VR_LST-LST10-V3.0] but

considering only version 3.0 and the corresponding in-situ matchups. Therefore, these results are shown to provide a qualitative assessment of the temporal consistency of version 3.0 through the years. Overall, results show a systematic underestimation (i.e., negative values) of the CLMS LST product relative to in-situ observations, with a larger bias in 2025, compared to previous years. The stability of the absolute residuals and RMSE between 2025 and previous years suggests that the apparent increase in the bias in 2025 is not due to a systematic product issue, but rather to differences in station sampling, as the overhaul bias is more sensitive to the inclusion or exclusion of stations than the other metrics.

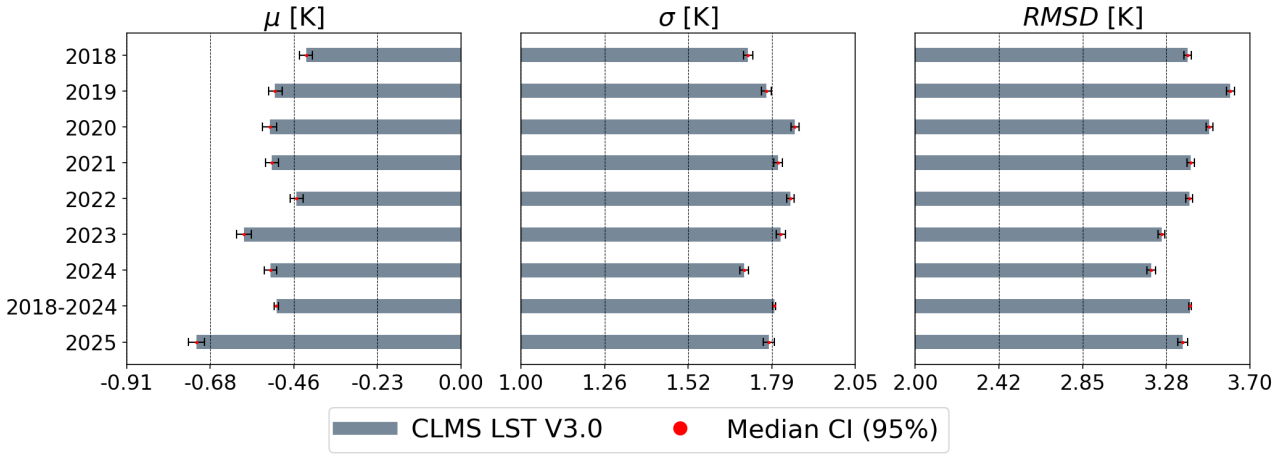


Figure 1 - Statistical summary of yearly metrics – bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD) – derived from hourly satellite/in-situ matchups in respect to CLMS LST version 3.0 product across all GBOV stations. Yearly statistics for 2018–2024, together with the full-period results, correspond to the analysis performed in the validation report [CGLOPS1_VR_LST-LST10-V3.0] but considering only matchups of version 3.0 against in-situ observations. Statistics for 2025 are shown in the last row. Error bars represent the 95 % confidence intervals of the median metric values. All values are expressed in Kelvin.

4.1.2 LST10-DC products

The LST 10-daily daily cycle (LST10-DC) family of products describes the properties of the diurnal cycle of LST during a dekad. Maximum, median, and minimum LST values for each hour are derived for each dekad from hourly LSTs, being then compared to the corresponding statistics derived from in-situ data. As in previous VRs, in-situ measurements within a 1-hour window around each cloudy satellite observation are excluded, ensuring that the resulting statistics are representative of clear-sky conditions. Summary statistics are provided for the LST10-DC maximum, median, and minimum components (Table 3, Table 4, and Table 5, respectively) for 2025, as well as for the period 2018–

2024 as reported in the validation report [CGLOPS1_VR_LST-LST10-V3.0] but considering only matchups of version 3.0 against in-situ observations.

For the LST10-DC maximum values (Table 3), version 3.0 shows an overall bias of about -0.674 ± 0.049 K, absolute residuals of 1.719 ± 0.042 K and RMSD of 3.166 ± 0.055 K for 2025. The bias is slightly higher in 2025 compared to the 2018–2024 period (from -0.41 K to -0.61 K), while absolute residuals and RMSD values remain within the overall range of variability observed during the assessment period (from 1.68 K to 1.86 K and from 3.08 K to 3.56 K, respectively). This behavior is consistent with the results obtained for the hourly LST product. The full 7-year statistics indicate a bias of -0.514 ± 0.015 K, absolute residuals of 1.789 ± 0.011 K, and RMSD of 3.350 ± 0.016 K.

The statistical summary for the LST10-DC median values (Table 4) highlights the behavior of the most representative component within the LST10-DC family of products. The median LST is less sensitive to extreme values than the maximum and minimum components and therefore provides a more robust representation of the dataset performance. For 2025, the median component presents an overall bias of -0.708 ± 0.049 K, absolute residuals of 1.787 ± 0.041 K, and RMSD of 3.104 ± 0.050 K. These values remain within the range of variability observed during the previous assessment period (2018–2024), particularly in terms of precision and RMSD. From 2020 to 2024, RMSD values show a gradual decrease from 3.29 K to 3.06 K, while absolute residuals decrease from 1.86 K to 1.76 K, indicating improved overall agreement between satellite and in-situ observations over time. The 2025 RMSD also remains lower than the full-period value for 2018–2024 (3.204 ± 0.015 K), although the latter includes a significant larger number of matchup samples (1700771 vs. 14250). In contrast, the 2025 bias reaches a higher negative value than the range observed during previous years.

For the case of LST10-DC minimum values (Table 5), the statistical summary shows an overall increase in bias, absolute residuals, and RMSD relative to the maximum and median values. For 2025, the minimum component shows a total bias of -0.939 ± 0.064 K, absolute residuals of 2.007 ± 0.044 K, and RMSD of 3.796 ± 0.065 K. From 2018 to 2024, bias values range between -0.57 K and -0.73 K, while RMSD values vary between 3.56 K and 3.80 K. The full-period statistics for 2018–2024 indicate a bias of -0.644 ± 0.018 K, absolute residuals of 1.997 ± 0.012 K, and RMSD of 3.713 ± 0.020 K. The larger errors obtained for 2025 may reflect the reduced number of stations and matchups samples used in the present validation analysis, as well as the increased sensitivity of nighttime/minimum LST values to local surface heterogeneity and atmospheric effects.

Overall, the LST10-DC 2025 assessment indicates that the CLMS version 3.0 product maintains stable performance and remains consistent with previous reported results.

Table 3 – Statistical summary of yearly metrics – bias (μ), absolute residuals (σ), and root-mean-squared difference (RMSD) – for LST10-DC maximum values derived from satellite/in-situ matchups of the CLMS LST version 3.0 product across all GBOV stations. Yearly statistics for 2018–2024, together with the full period results, are retrieved from the analysis performed in the validation report [CGLOPS1_VR_LST-LST10-V3.0], considering only version 3.0 matchups to in-situ observations. For the full period 2018–2024, values are reported with 95 % confidence intervals in brackets. Statistics for 2025 are shown in the last row. All values are expressed in Kelvin.

Period	μ [K]	σ [K]	RMSD [K]	#matchups
2018	-0.41	1.68	3.37	21095
2019	-0.61	1.76	3.56	23025
2020	-0.56	1.85	3.47	27001
2021	-0.48	1.81	3.34	28025
2022	-0.41	1.86	3.33	28013
2023	-0.58	1.80	3.22	23111
2024	-0.57	1.73	3.08	20501
2018–2024	-0.514 (±0.015)	1.789 (±0.011)	3.350 (±0.016)	170771
2025	**-0.674** (±0.049)	**1.719** (±0.042)	**3.166** (±0.055)	**14250**

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Table 4 – Same as Table 3 but for LST10-DC Median.

Period	μ [K]	σ [K]	RMSD [K]	#matchups
2018	-0.47	1.68	3.23	21095
2019	-0.62	1.75	3.39	23025
2020	-0.57	1.86	3.29	27001
2021	-0.54	1.83	3.18	28025
2022	-0.44	1.82	3.18	28013
2023	-0.62	1.80	3.08	23111
2024	-0.58	1.76	3.06	20501
2018–2024	-0.542 (±0.015)	1.794 (±0.011)	3.204 (±0.015)	170771
2025	**-0.708** (±0.049)	**1.787** (±0.041)	**3.104** (±0.050)	**14250**

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Table 5 – Same as Table 3 but for LST10-DC Minimum.

Period	μ [K]	σ [K]	RMSD [K]	#matchups
2018	-0.57	1.88	3.74	21095
2019	-0.73	1.95	3.80	23025
2020	-0.65	2.09	3.80	27001
2021	-0.61	2.04	3.70	28025
2022	-0.57	2.05	3.72	28013
2023	-0.72	1.98	3.56	21111
2024	-0.68	1.94	3.65	20501
2018–2024	-0.644 (±0.018)	1.997 (±0.012)	3.713 (±0.020)	170771
2025	**-0.939** (±0.064)	**2.007** (±0.044)	**3.796** (±0.065)	**14250**

4.1.3 LST10-Syn products

This family of products characterizes the range of LSTs within a dekad, regardless of the time of day. Similarly to the LST10-DC product, the maximum, median, and minimum LST values are derived for each dekad, but using data from all available hourly timeslots. Since the maximum and minimum synthesis components are more susceptible to sampling uncertainties, making comparisons with in-situ observations more challenging, these fields are considered auxiliary products in the present assessment. Therefore, the median LST is treated as the primary synthesis product.

In addition to the synthesis components, a Thermal Conditions Index (TCI) is also distributed to place the LSTs products in a climatological context. However, the TCI is not evaluated in the present report, since its validation against in-situ observations would require relatively longer observational records. Further details on the TCI are provided in the main validation report [CGLOPS1_VR_LST-LST10-V3.0].

Summary statistics for LST10-Syn median product are presented in Table 6. For 2025, the synthesis median shows an overall bias of -1.147 ± 0.141 K, absolute residuals of 1.082 ± 0.117 K, and RMSD

of 2.150 ± 0.138 K. Overall, the obtained values for 2025 remain within the expected range of variability observed during the 2018–2024 assessment period. In particular, the 2025 absolute residuals and RMSD values are lower than those obtained in previous yearly assessments, despite the significantly reduced number of matchups. In contrast, the bias shows greater sensitivity to changes in sample size, showing a relatively higher value in 2025 compared to the 2018–2024 average.

Table 6 – Same as Table 3 but for LST10-Syn Median.

Period	μ [K]	σ [K]	RMSD [K]	#matchups
2018	-0.85	1.12	2.29	919
2019	-1.05	1.14	2.39	1001
2020	-0.99	1.21	2.26	1180
2021	-1.06	1.22	2.73	1231
2022	-0.82	1.21	2.17	1207
2023	-1.01	1.26	2.17	1002
2024	-0.93	1.19	2.19	896
2018–2024	-0.953 (±0.058)	1.198 (±0.031)	2.333 (±0.069)	7436
2025	**-1.147** (±0.141)	**1.082** (±0.117)	**2.150** (±0.138)	**618**

4.2 TEMPORAL STABILITY

4.2.1 ERA5 reanalysis

Temporal stability and spatial consistency assessments of the CLMS LST version 3.0 product were previously performed in the main validation report [CGLOPS1_VR_LST-LST10-V3.0] using independent datasets, namely ERA5 skin temperature (SKT) reanalysis. The analysis covered the full reprocessing period (2018–2025) and was carried out at both global scale and across each individual GEO domains (GOES-16, MSG prime + IODC, and HMWR). Since the CLMS LST version 3.0 dataset for 2025 was already included in the comprehensive temporal stability assessment of the main

validation report, the present report provides only a summary of the main findings. Future VR updates will extend this evaluation as new yearly datasets become available.

Based on monthly mean LST anomaly differences, the ERA5 SKT comparison showed that version 3.0 exhibits improved temporal consistency relative to the previous operational version 2.2, with systematic lower anomaly differences, reduced variability, and weaker temporal trends (*Figure 12 of* CGLOPS1_VR_LST-LST10-V3.). Results over 2018–2025 highlighted temporal inconsistencies in version 2.2. associated with differences in input configuration throughout the operational period, since important updates – such as the introduction of dynamic vegetation-based surface emissivity and the integration of IODC data – were only implemented 2021 onward. In contrast, these updated inputs are consistently implemented in version 3.0 throughout the entire reprocessing period from 2018 onward, removing the temporal discontinuities previously identified in the GOES-16 and HMWR components. More details concerning the updated inputs implemented in CLMS LST version 3.0 are available in the product user manual [CGLOPS1_PUM_LST-V3.0]. At the global scale, mean anomaly differences remained within ± 0.2 K until the end of 2022, while trend magnitudes remained below the CLMS threshold requirement of 0.3 K per decade for all domains.

5 CONCLUSIONS

At the time of this report, several GBOV in-situ datasets were not available for 2025 or were reported as affected by inconsistencies in the LST reference measurements. This includes stations that have previously demonstrated high-quality and stable time-series in earlier validation reports, such as *Manhattan*, *Des Moines*, *Williams*, *DukeForest*, *Southern Great Plains*, and *Table Mountain*. A dedicated assessment and correction of these datasets is currently ongoing within GBOV, and updated records are expected to be made available in the near-future, enabling their inclusion in subsequent VRs. Since September 2023, GBOV has released version 2.2 of its in-situ reference measurements, which includes general corrections to existing LST datasets, including historical time-series, as well as the addition of new stations across the United States, Europe, Africa, Australia, and Asia. Continued improvements in data quality, temporal consistency, and station coverage are expected in future GBOV releases, further strengthening their use for satellite validation analysis.

This report confirms that the CLMS LST family of products version 3.0 maintain their consistency for the 2025 dataset, despite the lower number of satellite/matchup samples relative to previous yearly assessments. The LST hourly product is characterized by an overall bias of about -0.72 K, absolute residuals of 1.78 K, and RMSD of 3.36 K. Conformity tests show that approximately 50 % of all hourly matchups conform to the threshold requirement (i.e., below or equal to 2 K) while 27 % conform to the goal requirement (i.e., below or equal to 1 K). It is worth noting that, despite the use of less stations in the present report in comparison to the ones used for previous assessment periods (2018–2024), the obtained 2025 statistics are shown to be comparable.

The same conclusions can be applied to the LST10 family of products, which were evaluated using the corresponding statistics derived from clear-sky only in-situ data. Overall, the LST10-DC products show consistent performances relative to previous yearly assessments. For 2025, lower RMSD values are found for the median LST product (about 3.10 K), while higher errors are found for the minimum LST product (about 3.80 K). In term of bias, the maximum LST product is shown to provide a lower value (about -0.67 K) than median and minimum LST. Regarding the LST10-Syn product, only the median LST is evaluated in this report, since the maximum and minimum components are more susceptible to sampling uncertainties. Results showed an overall bias of approximately -1.15 K, absolute residuals of 1.08 K, and RMSD of 2.15 K, which are in good agreement with those obtained in previous yearly assessments, despite the lower number of matchup samples used.

The last part of this evaluation is focused on the temporal stability of the CLMS LST product. Since this assessment had been already performed in the main validation report [CGLOPS1_VR_LST-LST10-V3.0] over the full reprocessing period (2018–2025), for the present report only a summary of the obtained results was presented. In comparison to previous year performances, the results for 2025 showed consistent temporal behaviors of version 3.0 for the global and GEO domains.

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